

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
 B. Sc. Engineering 1st Year 2nd Term Backlog Examination, 2023
CE 1201
 (Engineering Mechanics)

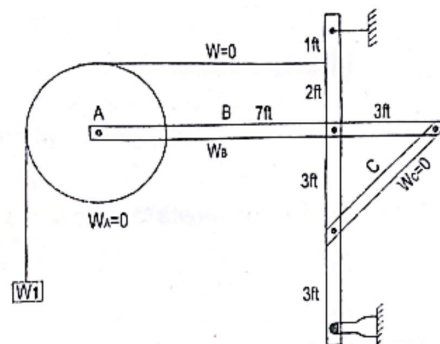
Full Marks: 210

Time: 3 hrs

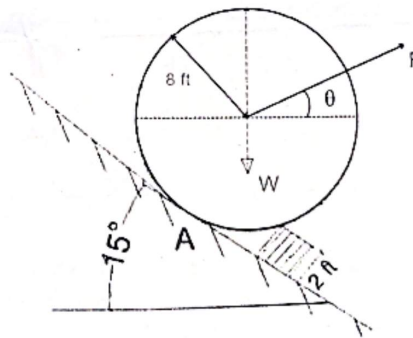
- N.B.** i) Answer any three questions from each section in separate script.
 ii) Figures in the right margin indicate full marks.
 iii) Assume reasonable value for any missing data.

Section – A

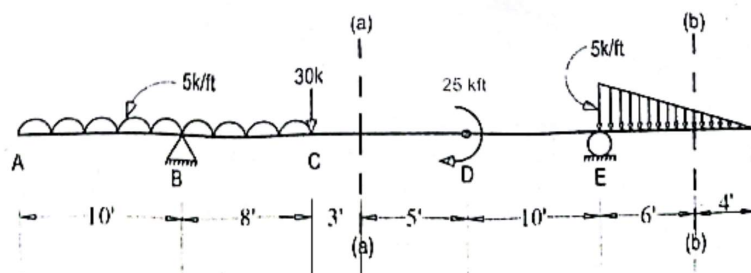
1. (a) Define - (i) Transmissibility of force, (ii) Two-force member, (iii) Free body diagram (18)
 and (iv) Equilibrium. Draw the free-body diagram of members of the following structure.



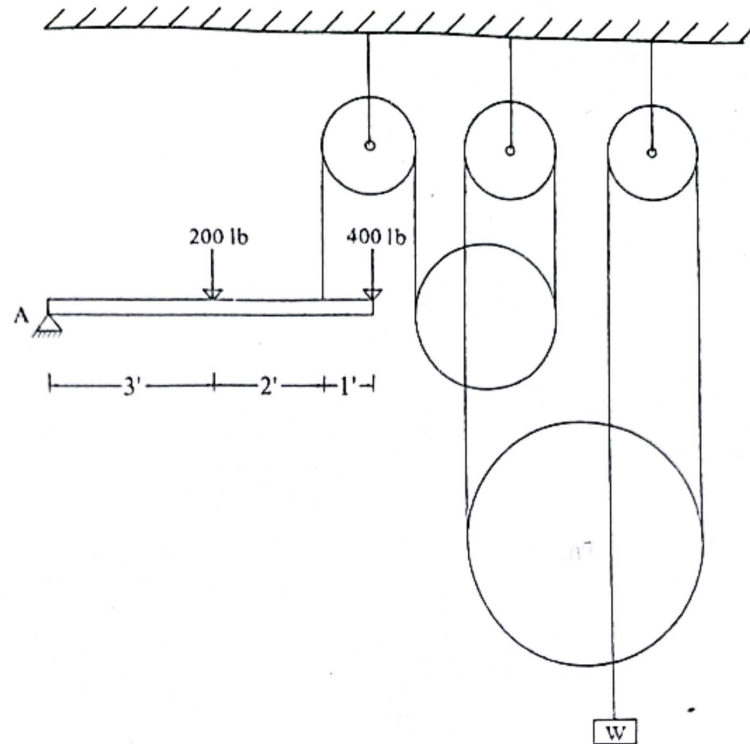
- (b) The wheel shown in figure is on the point of rolling over the block. If $W=1000$ lb, what (17)
 is the magnitude and sense of the least force F that will produce the condition.



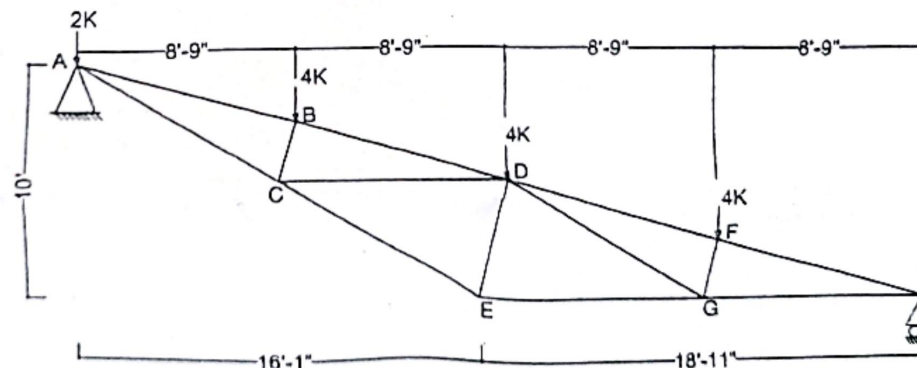
2. (a) Define shear force and bending moment. A beam is loaded as shown in figure below. (20)
 Find the reaction at A and E. Also calculate the shear force and bending moment at section (a)–(a) and (b)–(b).



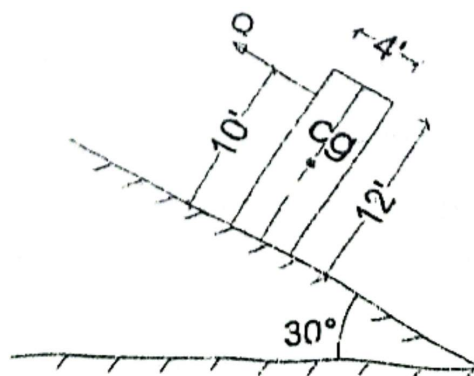
- (b) In the following figure below, what will be the value of 'W' if equilibrium exists? (15)
Assume the pulleys are frictionless.



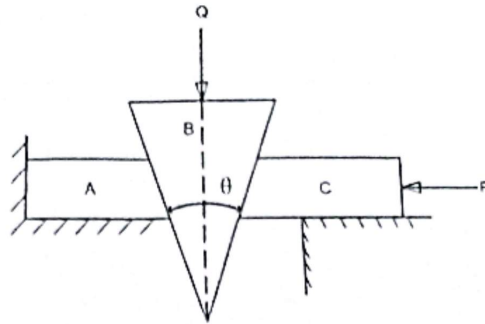
3. (a) For belt friction, prove that, $T_1 = T_2 e^{f\theta}$, where the symbols have their usual meanings. (15)
(b) Determine the force of the members located to the right of DE for the inverted Howe roof truss. (20)



4. (a) A relatively tall and slender body be as shown in figure below with a gradually increasing load Q applied to it. If $W = 105 \text{ lb}$, $f = 0.3$ and the inclination of the plane is 30° , will the body slide or tip over? (15)



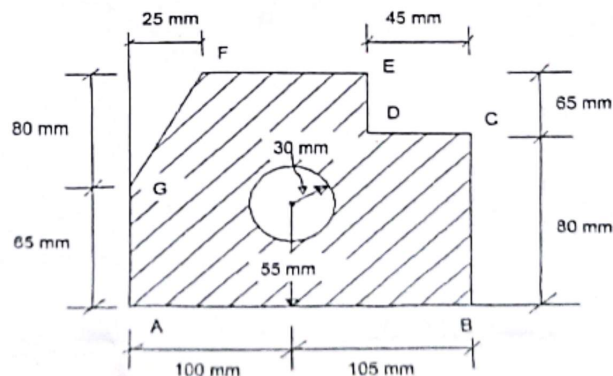
- (15)
- (b) A wedge B is inserted between a fixed surface A and a movable block C. Given that $W_A = 200 \text{ lb}$, $W_B = 500 \text{ lb}$, $W_C = 5000 \text{ lb}$, $\theta = 7.5^\circ$, and Force $R = 13000 \text{ lb}$. Consider $f = \frac{1}{3}$, what force Q will impose impending motion C. (15)



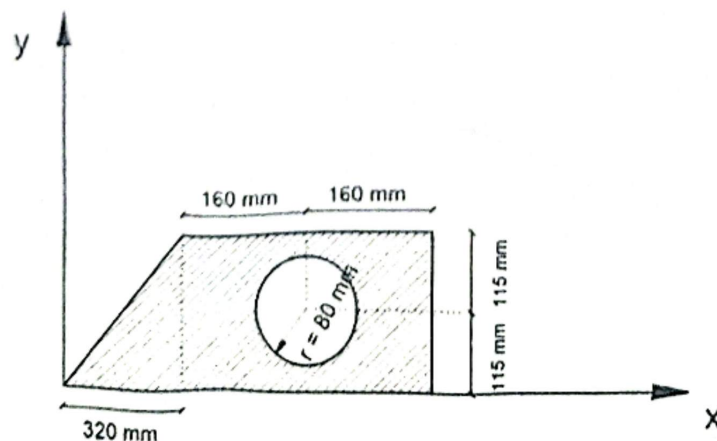
- (c) Distinguish between angle of friction and angle of repose. (05)

Section – B

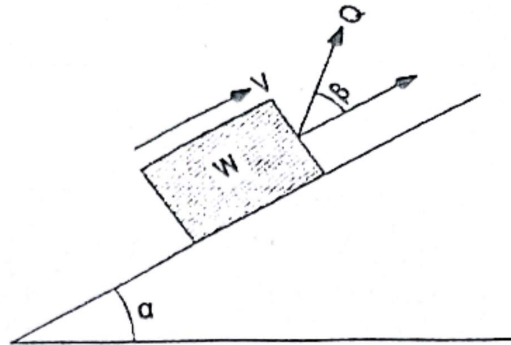
5. (a) What do you mean by centroid and center of gravity? Narrate the real life application of centroid. Find the centroid of a sector of a circle of radius r which subtends an angle of 2β . (15)
- (b) Determine the first moment of area about X and Y axis of the shaded area shown in figure below. (20)



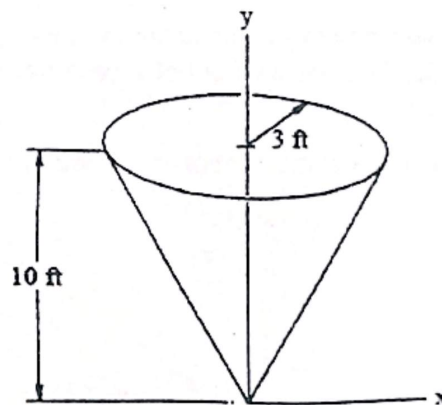
6. (a) State real life application of moment of inertia of area in construction engineering sector. (05)
- (b) Define the following factors: Moment of inertia, Polar moment of inertia and Radius of gyration. Prove that, $J = I_x + I_y$, where the symbols bear their usual meanings. (10)
- (c) Calculate the polar moment of inertia and radius of gyration with respect to X and Y axis of the shaded area as shown in figure. (20)



7. (a) State the principle of linear impulse and momentum. Prove that, $R\Delta t = \Delta mv$, where the symbols bear their usual meanings. (16)
- (b) In figure, $W = 200$ lb, $\alpha = 35^\circ$, $Q = 125$ lb, $\beta = 25^\circ$, $f = 1/3$ and the body is displaced at 12 ft up the inclined surface. Find the resultant force on W and the net work done. (19)



8. (a) Determine the volume and amount of paint required for the inside and outside surfaces of the cone and doughnut using Pappus-Guldinus theorem. It is noted that one gallon of paint covers 300 ft^2 . (13)



- (b) Distinguish between angle of friction and angle of repose. (08)
- (c) A 90 lb body is on a horizontal plane, where the co-efficient of kinetic friction, $f = 0.25$. A "Q and F" are applied to the body as shown in figure. (i) What is the impulse of Q and F force in the horizontal direction during 2.5 sec? (ii) What is the net impulse on the body during the same time? (14)

